

Kaustav Mukherjee

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EDUCATION

Carnegie Mellon University (CMU) <i>Master of Science in Computer Vision (GPA 4.22/4.33)</i>	Pittsburgh, PA Aug. 2024 – Dec. 2025
National University of Singapore (NUS) <i>Bachelor of Engineering in Mechanical Engineering (GPA 4.64/5.00)</i>	Singapore, Singapore Aug. 2020 – May 2024

EXPERIENCE

AI Engineer <i>Kaliber Labs</i>	Jan. 2026 – Present San Francisco, CA
- Built a real-time multi-view 3D reconstruction pipeline from surgical video using TSDF fusion, driving a 2-day showcase resulting in a signed LOI with a Fortune 500 medical device company and a paper accepted at ECCV - Developed an auxiliary loss term for JEPA-style world models, improving baselines by 50% in low-data scenarios - Improved textual and aural emotion recognition models by architecting a multi-task learning (MTL) framework for categorical classification and regression, driving a 30% absolute jump in accuracy - Engineered a stateless microservice architecture with NATS and Redis for a multimodal conversational AI PoC	
AI Engineer Intern <i>Kaliber Labs</i>	May 2025 – Aug. 2025 San Francisco, CA
- Architected a modular robotic navigation system with ROS2 and Nav 2, integrating 6+ CV models with C++ behavior trees via shared memory and multithreading, and deployed on the LG Q9 robot and NVIDIA Isaac Sim - Reduced per-frame edge processing latency from 300ms to 150ms across multiple CV inference tasks - Led technical integration across 10+ team members and delivered 10+ live demos to investors and stakeholders	
Software Engineering Intern <i>Kaliber Labs</i>	Jan. 2023 – Jul. 2024 San Francisco, CA
- Accelerated convergence of a surgical camera pose estimation algorithm by 10 times using particle swarm optimization (PSO) to match semantic segmentation masks between the scope view and a digital twin - Utilized NeRFs for real-time camera registration, enabling geodesic measurements with 90% accuracy - Fine-tuned stable diffusion to generate synthetic training data, expanding surgical tool datasets by up to 5 times - Employed artificial data augmentation to train multiple semantic segmentation models to above 97% accuracy - Optimized CV model inference pipelines via ONNX and quantization, reducing memory footprint by 40%	

PUBLICATIONS

Zhu, P., Penachio, S., Mukherjee, K., Jonelagadda, A. "Spectral-Target Physical Latent Structuring for JEPA-Style World Models" <i>arXiv</i> , 2026
Bychkov, D., Shah, R., Mukherjee, K., So, M. "Surface Coverage Completeness for Quality-Assured Arthroscopic 3D Reconstruction" <i>Proceedings of the ECCV Workshop on DCA-MI</i> , 2026 (To appear)
Riaz, S., Zheng, H., Roche, A., Zhang, M., Ragu, A. Penachio, S., Mukherjee, K., Jonelagadda, A. "Multi-Task Learning for Non-Canonical Phoneme Recognition via Articulatory Feature Decomposition" <i>arXiv</i> , 2026

PROJECTS

Bundle Adjustment for 360° Pose Correction <i>Gaussian Splatting, SfM, SLAM</i>	Aug. 2025 – Nov. 2025
Implemented a rig-based Structure from Motion (SfM) pipeline with bundle adjustment for Stella-VSLAM outputs from 360° images, improving Gaussian Splatting reconstruction quality by 10.5%	
Drone Perception System at CMU Airlab <i>Python, OpenCV, ROS, Computer Vision</i>	Aug. 2024 – Nov. 2024
Designed a depth-aware perception system for a drone with a 2-DOF manipulator using ROS and RGBD cameras	

SKILLS

Competencies: 3D Reconstruction, Computer Vision, SLAM, SfM, NeRF, Gaussian Splatting, Diffusion, World Models, Reinforcement Learning, LLM Fine-tuning, Context Engineering, Model Quantization
Frameworks: PyTorch, Tensorflow, NumPy, OpenCV, ROS2
Languages and Tools: Python, C++, JavaScript, Docker, Redis, NATS, ONNX, AWS, Linux, Git